

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	(d)	(i)		White solid / precipitate (of magnesium hydroxide)		1		1		1
		(ii)		Number of mol of MgS = $\frac{0.224}{56.4}$ = 0.00397 $\therefore$ 0.00397 mol of H ₂ S also produced (1) Volume of H ₂ S = 0.00397 $\times$ 24.0 = 0.095(3) dm ³ (1) = 95.(3) cm ³ (1) ecf possible Accept alternative method using pV = nRT	1	2		3	1	
	(e)			A colourless solution (1) The solubility of the group 2 hydroxides increases down the group (1)	1	1		2		1
	(f)			BaO + H ₂ O $\rightarrow$ Ba(OH) ₂ (1) pH > 7 (1)	2			2		
	(g)			Any two of following for (1) each - Barium is in group 2 and has two outer electrons - Too much energy is needed to remove a third electron - This necessitates removing an electron from a shell nearer to the nucleus (to produce a Ba³⁺ ion)	1	1		2		
				Question 8 total	9	7	0	16	2	2